

Mizuho Quant Analytics Workshop

Yield Curves & Swap Pricing

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MHI

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Global Quant Analytics

- MLIB is the Global Quant Analytics library at Mizuho
- Co-Developed by Mizuho Tokyo, London and New York

Features

- Yield Curves with CSA Support
- Swap Pricing & Risk
- Bond Pricing: US Treasuries, German Bunds, Japanese JGBs ...
- Credit Default & Total Return Swaps
- IR Options: Swaptions, Caps, Floors ...
- Exotic Options: IR / FX Hybrids

- ① Getting Started
- ② Developer Tools
- ③ MLIB Ribbon
- ④ MLIB Functionality
- ⑤ Holiday Calendars
- ⑥ Daycount Conventions
- ⑦ Date Functions
- ⑧ Date Schedules
- ⑨ Utility Functions
- ⑩ Yield Curves, Forward Rates & Discount Factors
- ⑪ Swap Pricing & Risk
- ⑫ Swaps Pricer Tool

Excel Launcher

X:/Secure/DIR_MLIB_Analytics/MLIB_Excel_Addin

Features

- Function groups for navigation ease: meCurve, meProducts ...
- Multi Yield Curve construction & OIS Discounting with CSA Support
- OIS, IRS, Xccy and Asset Swap Pricing and Risk
- Every function has an Excel Example Template

Documentation

X:/Secure/DIR_MLIB_Analytics/Documentation

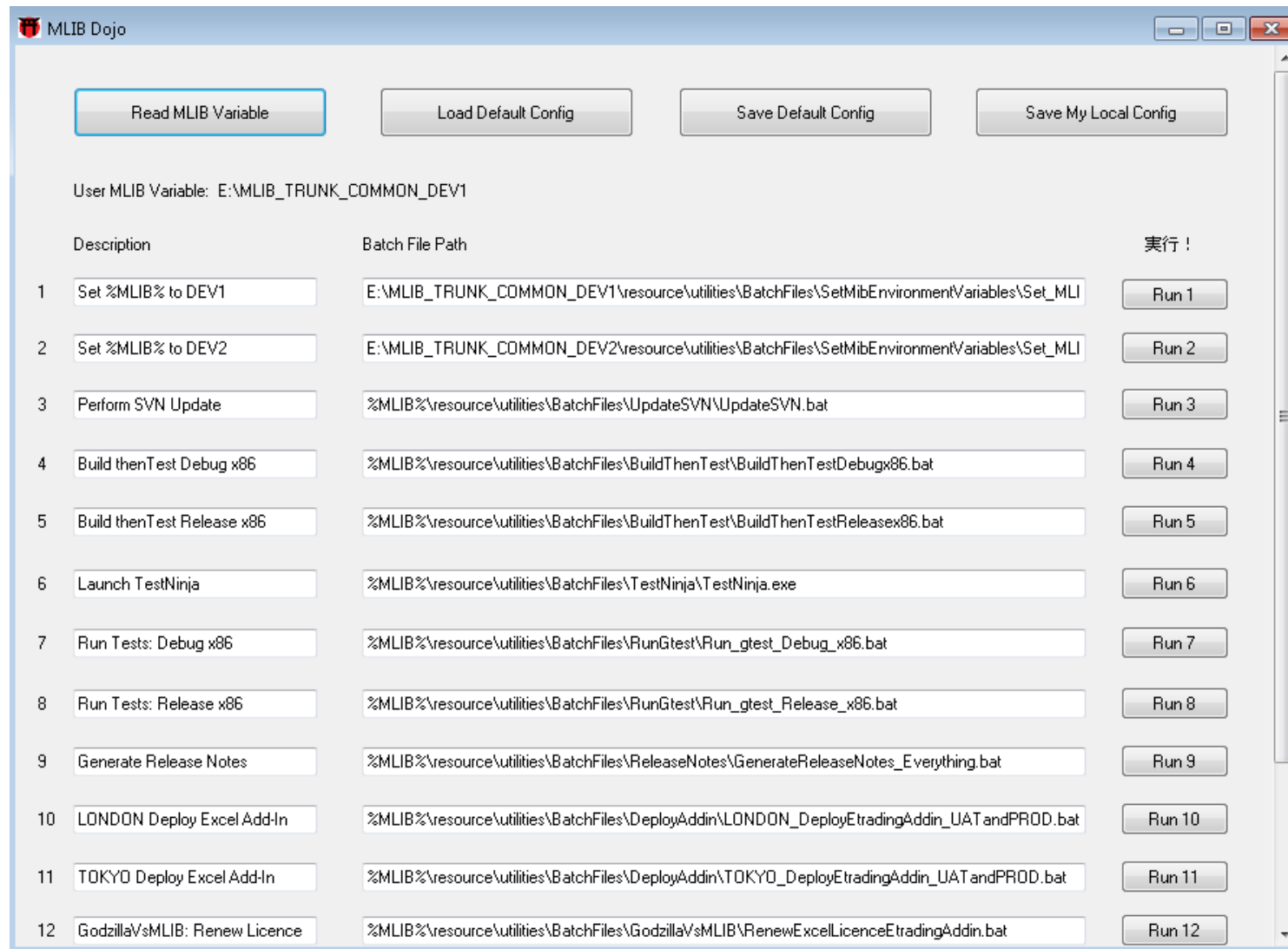
Examples

X:/Secure/DIR_MLIB_Analytics/Examples

Workshops

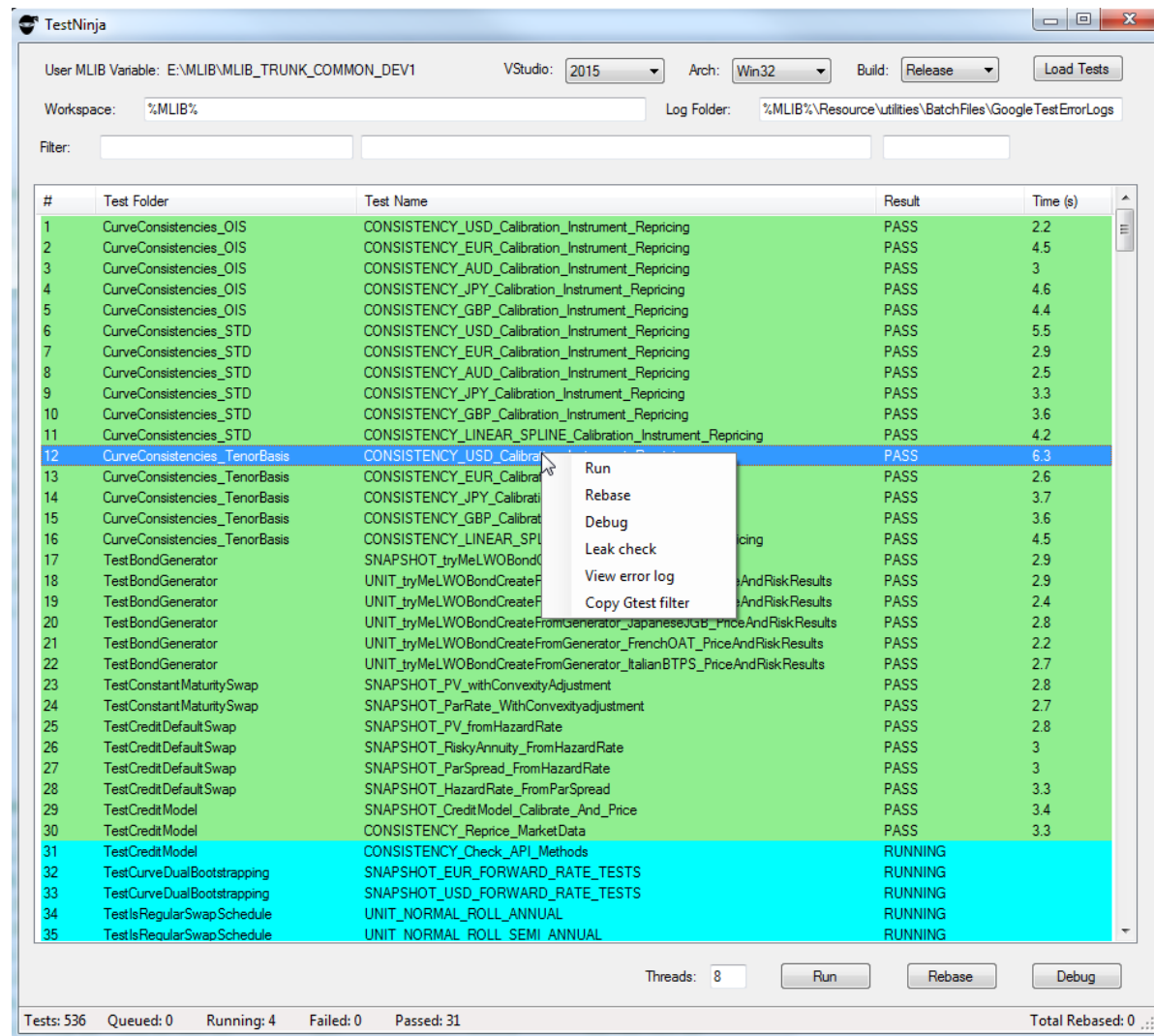
X:/Secure/DIR_MLIB_Analytics/Workshops

Automation Dashboard - On Demand Library Updates

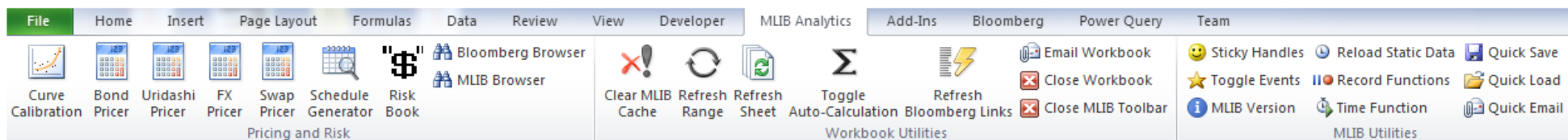


Test Harness

- Before every code change 6,000+ price safety tests must pass



MLIB Analytics Ribbon



Highlights

Curve Calibration

Quant Curve Calibration Tool - Used to confirm all major curve currencies and their instruments reprice

Bond Pricer

Bond Pricing Tool - Used to confirm all Bonds in the following markets reprice vs Bloomberg. Supports government bonds for : USA, Germany, UK, Japan, France, Italy, Holland, Australia et al

Swap Pricer

Live Swap Pricing and Risk Templates for: Asset Swaps, OIS, IRS, Tenor Basis, Xccy Basis, Variable Notional Swaps et al.

Clear MLIB Cache

Clear all objects stored in the MLIB Memory Cache

Refresh Bloomberg Links

Refresh Static and Non-Volatile Bloomberg Functions

Sticky Handles

Toggle the Object Safety Feature On / Off (discussed later)

MLIB Version

Show the MLIB Version and License Expiry Date!!!

Record MLIB Functions

Record / Log MLIB Functions

Quick Save, Load and Email

Highlighting an LWO object and clicking these buttons will quickly save, load and email the object respectively, using the C:\Temp folder

MLIB Functions (not an exhaustive list)

meDate...

meDateBusinessDays
meDateCentralBank
meDateECB
meDateFromTenor
meDateFromYearFraction
meDateFuturesContract
meDateIMMFromMonth
meDateIsHoliday
meDateIsWeekday
meDateIsWeekend
meDateIsWorkingDay
meDateNthIMM
meDateNthIMMFromStartDate
meDateYearFraction

meUtilities...

meUtilityClearLWOCache
meUtilityLVB
meUtilityResize
meUtilityRecord
meUtilityReplay
meUtilityVersion

meCurve...

meCurveCalibrateOIS
meCurveCalibrateSwap
meCurveCalibrateBasis
meCurveCalibrateFXForwards
meCurveDiscountFactors
meCurveForwardRates

meLWO...

meLWOExists
meLWOType
meLWOList
meLWODelete
meLWODeleteAll
meLWOSave
meLWOLoad

meLWOSchedule

meLWOScheduleCreate
meLWOScheduleCreateBespoke
meLWOScheduleDisplay

meLWOCurve...

meLWOCurveCreateOIS
meLWOCurveCreateSwap
meLWOCurveCreateBasis
meLWOCurveCreateFXForwards
meLWOCurveDiscountFactors
meLWOCurveForwardRates
meLWOCurveCalibrate
meLWOCurveMarketDataCreate
meLWOCurveMarketDataDisplay
meLWOCurveDisplay
meLWOCurveGeneratorCreate
meLWOCurveGeneratorDisplay

Functions groups used for navigational ease

meLWOSwap...

meLWOSwapCreate
meLWOSwapCreateFromGenerator
meLWOSwapAddFee
meLWOSwapDisplayCashflows
meLWOSwapDisplay
meLWOSwapAnnuity
meLWOSwapParRate
meLWOSwapParSpread
meLWOSwapPV01
meLWOSwapDV01
meLWOSwapDeltaLadder

meLWOBond...

meLWOBondCreate
meLWOBondDirtyPrice
meLWOBondCleanPrice
meLWOBondAccruedInterest
meLWOBondAccruedInterestDays
meLWOBondDisplaySchedule
meLWOBondDisplayCashflows
meLWOBondYield
meLWOBondCompoundYield
meLWOBondModifiedDuration
meLWOBondDV01
meLWOBondDV01Numerical

meLWOFixingTable

meLWOFixingTableCreate
meLWOFixingTableDisplay
meLWOFixingTableValues

meLWOXccy

meLWOXccySwapCreate
meLWOXccySwapParSpread
meLWOXccySwapSpread

meLWOFee

meLWOFeeLegCreate
meLWOFeeScheduleCreate

Holiday Data Sourced from SwapsWire

- | | | |
|--|---|--|
| ☐ ADB (Adelaide) | ☐ HKB (Hong Kong) | ☐ PAB (Paris) |
| ☐ AEB (AbuDhabi) | ☐ HKS (Hong Kong Stock Exchange trading) | ☐ PEB (Perth) |
| ☐ AMB (Amsterdam) | ☐ HAB (Hanoi) | ☐ PRB (Prague) |
| ☐ ANB (Ankara) | ☐ HEB (Helsinki) | ☐ RJB (Rio de Janeiro) |
| ☐ ATB (Athens) | ☐ HMB (Hamilton(Bermuda)) | ☐ RIB (Riyadh) |
| ☐ AUB (Auckland) | ☐ HOB (Hobart) | ☐ RKB (Reykjavik) |
| ☐ BAB (Beunos Aires) | ☐ ISB (Istanbul) | ☐ SOM (Eurex trading (German equity contracts)) |
| ☐ BMA (US Treasuries (SIFMA recommended close)) | ☐ JAB (Jakarta) | ☐ SON (Eurex trading (STOXX contracts)) |
| ☐ BBB (Brisbane) | ☐ JOB (Johannesburg) | ☐ SPB (Sao Paulo) |
| ☐ BCB (Bucharest) | ☐ KLB (Kuala Lumpur) | ☐ SAB (Santiago) |
| ☐ BDB (Budapest) | ☐ LAB (Labuan) | ☐ SEB (Seoul) |
| ☐ BEB (Beijing) | ☐ LBB (Lubljana) | ☐ SHB (Shanghai) |
| ☐ BKB (Bangkok) | ☐ LMB (Lima) | ☐ SIB (Singapore) |
| ☐ BMB (Mumbai) | ☐ LNB (London) | ☐ SOB (Sofia) |
| ☐ BOB (Bogota) | ☐ LSB (Lisbon) | ☐ STB (Stockholm) |
| ☐ BRB (Brussels) | ☐ LXB (Luxembourg) | ☐ SXB (Stuttgart) |
| ☐ BTB (Bratislava) | ☐ MDB (Madrid) | ☐ SYB (Sydney) |
| ☐ BXB (Brasilia) | ☐ MEB (Melbourne) | ☐ TYO (ISDA Standard TYO) |
| ☐ CMF (Chicago Mercantile Exch. trading (Equity)) | ☐ MIB (Milan) | ☐ TGT (TARGET) |
| ☐ CHS (Chicago Stock Exchange trading) | ☐ MNB (Manila) | ☒ TKB (Tokyo) |
| ☐ CNB (Canberra) | ☐ MOB (Montreal) | ☐ TPB (Taipei) |
| ☐ COB (Copenhagen) | ☐ MWB (Moscow) | ☐ TRB (Toronto) |
| ☐ DEL (Delaware) | ☐ MXB (Mexico City) | ☐ VIB (Vienna) |
| ☐ DBB (Dubai) | ☐ NBR (Brazilian payments system) | ☐ WAB (Warsaw) |
| ☐ DRB (Darwin) | ☐ NYB (New York) | ☐ WEB (Wellington) |
| ☐ DUB (Dublin) | ☐ NYS (New York Stock Exchange trading) | ☐ ZAB (Zagreb) |
| ☐ FRB (Frankfurt) | ☐ NDQ (NASDAQ) | ☐ ZUB (Zurich) |
| ☐ GEB (Geneva) | ☐ OSB (Oslo) | |

BusinessDayAdjustment

MOD_FOLLOWING

FOLLOWING

MOD_PRECEDING

PRECEDING

NO_CHANGE

Daycount

30/360

ACT/360

ACT/365FJ

ACT/365

ACT/365_ISDA

ACT/ACT

ACT/ACT_ISMA

Date Functions

meDate...

[meDateBusinessDays](#)

[meDateCentralBank](#)

[meDateECB](#)

[meDateFromTenor](#)

[meDateFromYearFraction](#)

[meDateFuturesContract](#)

[meDateIMMFromMonth](#)

[meDateIsHoliday](#)

[meDateIsWeekday](#)

[meDateIsWeekend](#)

[meDateIsWorkingDay](#)

[meDateNthIMM](#)

[meDateNthIMMFromStartDate](#)

[meDateYearFraction](#)

meDateBusinessDays

From	Tue, 6-Jun-17
To	Tue, 13-Jun-17
Calendar	TGT
NumberOfBusinessDays	5

meDateFromTenor

StartDate	Tue, 6-Jun-17
Tenor	6M
BusinessDayAdjustment	MOD_FOLLOWING
Calendar	TGT
ResultDate	Wed, 6-Dec-17

meDateFromYearFraction

StartDate	Tue, 6-Jun-17
YearFraction	0.5
DayCount	ACT/365
ResultDate	Wed, 6-Dec-17

meDateFuturesContract

Ticker	Z9
ResultDate	Wed, 18-Dec-19

meDateIMMFromMonth

Year	2017
Month	3
Calendar	NYB
BusinessDayAdjustment	MOD_FOLLOWING
ResultDate	Wed, 15-Mar-17

meDateCentralBank

CentralBankID	ECB
ReferenceDate	Tue, 6-Jun-17
NextCentralBankDate	Thu, 8-Jun-17

meDateECB

ReferenceDate	Tue, 6-Jun-17
NextECBDate	Thu, 8-Jun-17

Swap Schedule Generation

meProductSwapSchedule

EffectiveDate	06-Jun-17
MaturityDate	2Y
AccrualDaycount	ACT/360
AccrualFrequency	Quarterly
AccrualBusinessDayAdjustment	MOD_FOLLOWING
AccrualCalendar	TGT
PaymentFrequency	Quarterly
PaymentBusinessDayAdjustment	MOD_FOLLOWING
PaymentCalendar	TGT
PaymentLag	0D
FixingBusinessDayAdjustment	MOD_FOLLOWING
FixingCalendar	TGT
FixingLag	0D
FixingAdvanceOrArrear	ADVANCE
FirstStubDate	
LastStubDate	
RollDay	IMM

FixingDate	AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate
Tue, 6-Jun-17	Tue, 6-Jun-17	Wed, 21-Jun-17	15	0.0417	Wed, 21-Jun-17
Wed, 21-Jun-17	Wed, 21-Jun-17	Wed, 20-Sep-17	91	0.2528	Wed, 20-Sep-17
Wed, 20-Sep-17	Wed, 20-Sep-17	Wed, 20-Dec-17	91	0.2528	Wed, 20-Dec-17
Wed, 20-Dec-17	Wed, 20-Dec-17	Wed, 21-Mar-18	91	0.2528	Wed, 21-Mar-18
Wed, 21-Mar-18	Wed, 21-Mar-18	Wed, 20-Jun-18	91	0.2528	Wed, 20-Jun-18
Wed, 20-Jun-18	Wed, 20-Jun-18	Wed, 19-Sep-18	91	0.2528	Wed, 19-Sep-18
Wed, 19-Sep-18	Wed, 19-Sep-18	Wed, 19-Dec-18	91	0.2528	Wed, 19-Dec-18
Wed, 19-Dec-18	Wed, 19-Dec-18	Wed, 20-Mar-19	91	0.2528	Wed, 20-Mar-19
Wed, 20-Mar-19	Wed, 20-Mar-19	Thu, 6-Jun-19	78	0.2167	Thu, 6-Jun-19

Bond Schedule Generation

meProductBondSchedule

EffectiveDate	22-May-18
AccrualFrequency	Semi-Annual
PaymentFrequency	Semi-Annual
FixedRate	2.000%
AccrualDayCount	Act/Act
AccrualBusinessDayAdjustment	NO_CHANGE
AccrualCalendar	NYB
PaymentBusinessDayAdjustment	NO_CHANGE
PaymentCalendar	NYB
PaymentLag	0D
MaturityDate	15-Nov-26

Fixed Bond Schedule

AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate
Tue, 22-May-18	Thu, 15-Nov-18	177	0.4849	Thu, 15-Nov-18
Thu, 15-Nov-18	Wed, 15-May-19	181	0.4959	Wed, 15-May-19
Wed, 15-May-19	Fri, 15-Nov-19	184	0.5041	Fri, 15-Nov-19
Fri, 15-Nov-19	Fri, 15-May-20	182	0.4976	Fri, 15-May-20
Fri, 15-May-20	Sun, 15-Nov-20	184	0.5027	Sun, 15-Nov-20
Sun, 15-Nov-20	Sat, 15-May-21	181	0.4955	Sat, 15-May-21
Sat, 15-May-21	Mon, 15-Nov-21	184	0.5041	Mon, 15-Nov-21
Mon, 15-Nov-21	Sun, 15-May-22	181	0.4959	Sun, 15-May-22
Sun, 15-May-22	Tue, 15-Nov-22	184	0.5041	Tue, 15-Nov-22
Tue, 15-Nov-22	Mon, 15-May-23	181	0.4959	Mon, 15-May-23
Mon, 15-May-23	Wed, 15-Nov-23	184	0.5041	Wed, 15-Nov-23
Wed, 15-Nov-23	Wed, 15-May-24	182	0.4976	Wed, 15-May-24
Wed, 15-May-24	Fri, 15-Nov-24	184	0.5027	Fri, 15-Nov-24
Fri, 15-Nov-24	Thu, 15-May-25	181	0.4955	Thu, 15-May-25
Thu, 15-May-25	Sat, 15-Nov-25	184	0.5041	Sat, 15-Nov-25
Sat, 15-Nov-25	Fri, 15-May-26	181	0.4959	Fri, 15-May-26
Fri, 15-May-26	Sun, 15-Nov-26	184	0.5041	Sun, 15-Nov-26

Featuring

- Version & License Expiry Information
- Recording, Replay & Logging
- Resizing Data Blocks
- Data Cleansing & Filtering
- Appending & Gluing Data

meUtilities...

meUtilityAppend
meUtilityClean
meUtilityClearLWOCache
meUtilityDataFilter
meUtilityLVB
meUtilityResize
meUtilityRecord
meUtilityReplay
meUtilityVersion

meUtilityClearLWOCache

1=meUtilityClearLWOCache()

meUtilityVersion

MLIB Addin: Version Number: 20180518.1629. Last built on May 18 2018 at 16:29:40. License will expire on Sep 1 2018

meUtilityRecord

meUtilityRecord(Enable=True\False, Folder="C:\Temp\")

- Data Table Resizing
- Removes unwanted #N/As

Input

Data	Value1	Value2	#N/A
1	100	10000	#N/A
2	200	40000	#N/A
3	300	90000	#N/A
4	400	160000	#N/A
5	500	250000	#N/A
6	600	360000	#N/A
7	700	490000	#N/A
8	800	640000	#N/A
9	900	810000	#N/A
10	1000	1000000	#N/A
#N/A	#N/A	#N/A	#N/A
#N/A	#N/A	#N/A	#N/A
#N/A	#N/A	#N/A	#N/A

meUtilityResize

Data	Value1	Value2
1	100	10000
2	200	40000
3	300	90000
4	400	160000
5	500	250000
6	600	360000
7	700	490000
8	800	640000
9	900	810000
10	1000	1000000

Label Value Blocks, LVBs

- Within MLIB Data Blocks are called Label Value Blocks
- Data Blocks can be appended or glued together

Input

Label1	Label2	Label3	Label4	Label5
Value1	Value2	Value3	Value4	Value5

meUtilityLVB

Label1	Value1
Label2	Value2
Label3	Value3
Label4	Value4
Label5	Value5

Input1

Label1	Value1
Label2	Value2

Input2

Label3	Value3
Label4	Value4
Label5	Value5

meUtilityLVBFromMultipleKeysValues

Label1	Value1
Label2	Value2
Label3	Value3
Label4	Value4
Label5	Value5

- **meUtilityClean**: Removes errors and blanks from data
- **meUtilityDataFilter**: Removes Duplicates from a data list

Input

1Y	1
2Y	2
3Y	#N/A
4Y	
5Y	5
10Y	10
20Y	20
25Y	#N/A
30Y	30

meUtilityClean

1Y	1
2Y	2
5Y	5
10Y	10
20Y	20
30Y	30

Input

Binance
Bitfinex
Bitfinex
HitBTC
HitBTC
Conabase GDAX
BitStamp

meUtilityFilter

Binance
Bitfinex
HitBTC
Conabase GDAX
BitStamp

Yield Curves

Technical Notes

- MLIB supports Multi-Curve Construction with CSA Support
- Yield Curves were discussed at length in our previous Workshop
- See X:/Secure/DIR_MLib_Analytics/Workshop/Workshops_2016-10
- For CSA Calculations we assume FX Forward Invariance

FX Forward Invariance Example for EUR_JPYCSA

$$FwdFX(t, T)^{EUR/JPY} = S \underbrace{\frac{P(t, T)^{EUR_USDCSA}}{P(t, T)^{JPY_USDCSA}}}_{\text{From Xccy Curve}} = S \underbrace{\frac{P(t, T)^{EUR_JPYCSA}}{P(t, T)^{JPY_JPYCSA}}}_{\text{JPY_JPYCSA from OIS Curve}}$$

where t denotes the valuation date, S the FX spot rate, $P(t, T)$ the discount factor at time t for tenor T with $0 \leq t \leq T$

CREATE EUROIS CURVE

MARKETDATAPROPERTIES

Currency	EUR
CurveType	OIS
CurveIndexFrequency	DAILY
AsOfDate	Tue, 22-May-18

Create Market Data Handle

Handle Name	EUR_OIS_CURVE_MARKETDATA
Market Data Handle	1 EUR_OIS_CURVE_MARKETDATA@23:3

Curve Dictionary

Handle Name	EUR_OIS_CURVE
Curve Generator	EUR_OIS
Domestic Curve Collection	EURYC
Foreign Curve Collection	

Curve Build

Handle Name	2 EUR_OIS_CURVE@29:3
Curve Collection	EURYC
Curve Index	OIS:EUROIS:EURDF:EURDF_EURCSA:EURDF_N
Calibration Time Taken	0.01700 sec
Curve Loaded	TRUE

OIS

Term	Rate	StartDate	EndDate	Use
ON	-0.36600%			TRUE
ECB1	-0.36100%	20-Jun-18	18-Sep-18	TRUE
ECB2	-0.36000%	01-Aug-18	30-Oct-18	TRUE
ECB3	-0.36000%	19-Sep-18	18-Dec-18	TRUE
ECB4	-0.35900%	31-Oct-18	29-Jan-19	TRUE
12M	-0.35400%			TRUE
18M	-0.31900%			TRUE
2Y	-0.25700%			TRUE

LIBOROISBASISSPREADS

Term	Rate
3Y	0.0840%
4Y	0.0930%
5Y	0.1020%
6Y	0.1080%
7Y	0.1120%
8Y	0.1140%
9Y	0.1150%
10Y	0.1160%
11Y	0.1160%
12Y	0.1160%
15Y	0.1150%
20Y	0.1100%
25Y	0.1060%
30Y	0.1030%
40Y	0.0960%
50Y	0.0890%

SWAPS

Term	Rate
3Y	-0.01400%
4Y	0.17000%
5Y	0.33600%
6Y	0.48800%
7Y	0.62500%
8Y	0.74800%
9Y	0.85900%
10Y	0.95800%
11Y	1.04500%
12Y	1.12400%
15Y	1.30600%
20Y	1.47300%
25Y	1.52900%
30Y	1.54600%
40Y	1.54400%
50Y	1.51200%

- Libor Forecast Rates do not depend on CSA Collateral

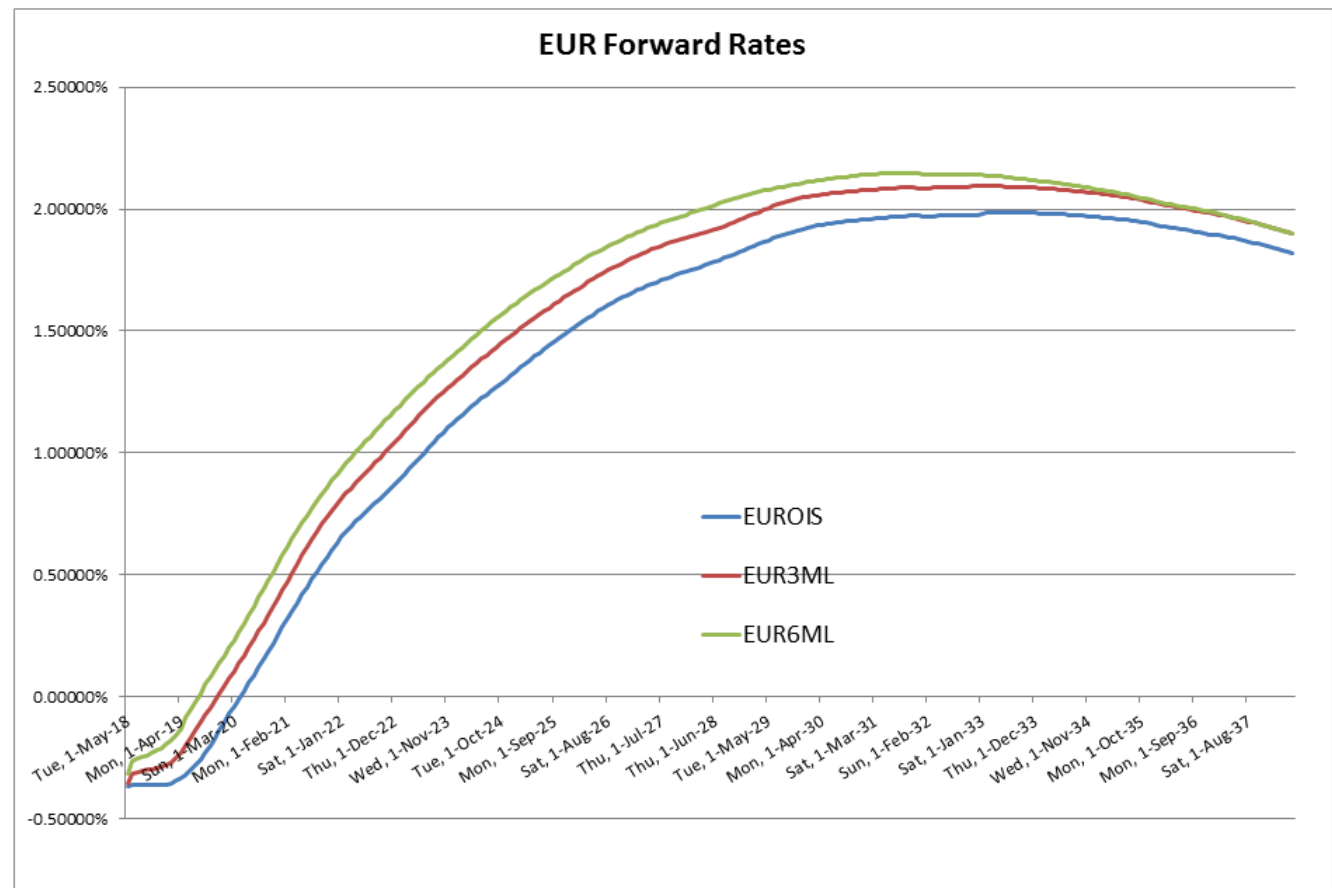
meCurveForwardRates

FixingDate	EUROIS	EUR3ML	EUR6ML
Tue, 22-May-18	-0.36600%	-0.35627%	-0.31458%
Fri, 22-Jun-18	-0.36100%	-0.31470%	-0.26573%
Mon, 23-Jul-18	-0.36100%	-0.30958%	-0.25424%
Thu, 23-Aug-18	-0.36000%	-0.30446%	-0.24626%
Mon, 24-Sep-18	-0.36000%	-0.29970%	-0.23800%

meLWOCurveForwardRatesTable

Start	0Y
End	20Y

Fixing Dates	EUROIS	EUR3ML	EUR6ML
Tue, 22-May-18	-0.36600%	-0.35627%	-0.31458%
Fri, 22-Jun-18	-0.36100%	-0.31470%	-0.26573%
Mon, 23-Jul-18	-0.36100%	-0.30958%	-0.25424%
Wed, 22-Aug-18	-0.36000%	-0.30463%	-0.24652%
Mon, 24-Sep-18	-0.36000%	-0.29970%	-0.23800%
Mon, 22-Oct-18	-0.36000%	-0.29672%	-0.22867%
Thu, 22-Nov-18	-0.35900%	-0.29342%	-0.21921%
Mon, 24-Dec-18	-0.35900%	-0.28872%	-0.20900%
Tue, 22-Jan-19	-0.35900%	-0.27789%	-0.19497%
Fri, 22-Feb-19	-0.35425%	-0.26632%	-0.17769%
Fri, 22-Mar-19	-0.34670%	-0.25430%	-0.15725%
Mon, 22-Apr-19	-0.33504%	-0.22671%	-0.13167%
Wed, 22-May-19	-0.31968%	-0.20000%	-0.08717%
Mon, 24-Jun-19	-0.29921%	-0.16689%	-0.04993%
Mon, 22-Jul-19	-0.27952%	-0.13725%	-0.01907%
Thu, 22-Aug-19	-0.25524%	-0.10516%	0.01511%
Mon, 23-Sep-19	-0.22764%	-0.07332%	0.05039%
Tue, 22-Oct-19	-0.20063%	-0.04523%	0.08236%
Fri, 22-Nov-19	-0.16863%	-0.01532%	0.11119%



- Discount Factors depend on CSA Collateral

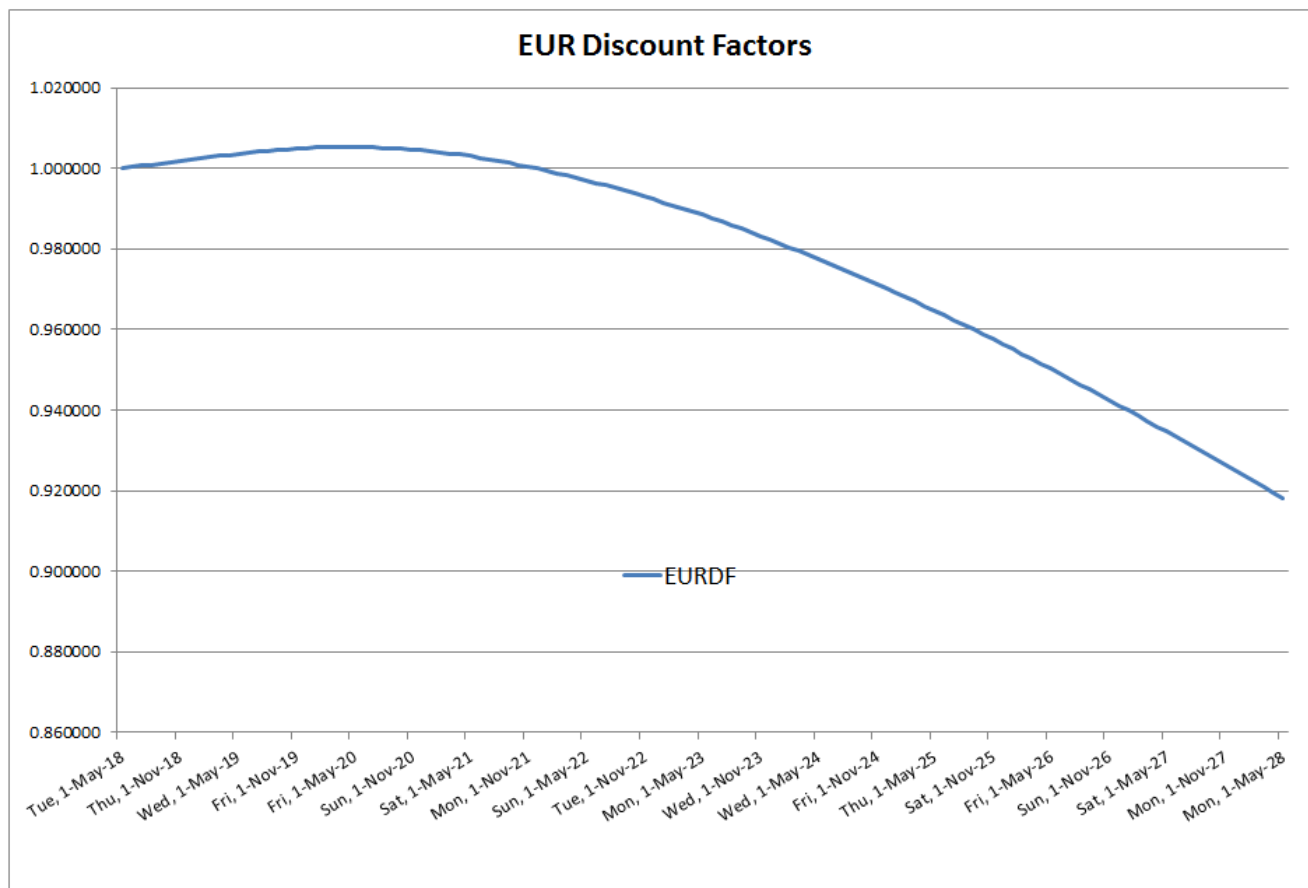
meCurveDiscountFactors

PaymentDate	EURDF
Tue, 22-May-18	1.000000
Fri, 22-Jun-18	1.000311
Mon, 23-Jul-18	1.000622
Thu, 23-Aug-18	1.000933
Mon, 24-Sep-18	1.001253

meLWOCurveDiscountFactorsTable

Start	OD
End	10Y

Payment Dates	EURDF
Tue, 22-May-18	1.000000
Fri, 22-Jun-18	1.000311
Sun, 22-Jul-18	1.000612
Wed, 22-Aug-18	1.000923
Sat, 22-Sep-18	1.001233
Mon, 22-Oct-18	1.001533
Thu, 22-Nov-18	1.001843
Sat, 22-Dec-18	1.002143
Tue, 22-Jan-19	1.002453
Fri, 22-Feb-19	1.002762
Fri, 22-Mar-19	1.003035
Mon, 22-Apr-19	1.003330
Wed, 22-May-19	1.003605
Sat, 22-Jun-19	1.003873
Mon, 22-Jul-19	1.004116
Thu, 22-Aug-19	1.004348
Sun, 22-Sep-19	1.004558
Tue, 22-Oct-19	1.004738
Fri, 22-Nov-19	1.004899
Sun, 22-Dec-19	1.005028



Swap Pricing & Risk

Swap Definition

LEGTYPE	Leg1:Fixed	Leg2:Float
FORECASTCURVE		EUR6ML
COMPOUNDMETHOD	NONE	GEOMETRIC
DISCOUNTCURVE	EUROIS	EUROIS
VALUATIONCURRENCY	EUR	EUR
CURRENCY	EUR	EUR
FIRSTFIXING		
LASTFIXING		
FIRSTSTUBCURVEINDEX		NATURAL
LASTSTUBCURVEINDEX		NATURAL
NOTIONAL	1,000,000	1,000,000
NOTIONALEXCHANGE	NONE	NONE
LEVERAGE	1	1
PAYRECEIVE	PAY	RECEIVE
EFFECTIVEDATE	24-May-18	24-May-18
MATURITYDATE	5Y	5Y
ACCRUALFREQUENCY	Annual	Semi-Annual
ACCRUALBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
ACCRUALCALENDAR	TGT	TGT
ACCRUALDAYCOUNT	30/360	ACT/360
PAYMENTFREQUENCY		
PAYMENTBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
PAYMENTCALENDAR	TGT	TGT
PAYMENTLAG	0D	0D
FIRSTSTUBDATE		
LASTSTUBDATE		
ROLLDAY	NORMAL	NORMAL
STUBTYPE	ShortStart	ShortStart

Create Swap

CreateSwap **MYEURSWAP@1:1**
Curve EURYC

Swap Pricing

PV **-29,334**
ParRate **0.41300%**
PV01 **500**
DV01 **509**

Generator Templates

- Quick Trade Entry for Standardized Products
- Supports Quick Save, Load, Email Features
- Can use Default Generator Templates or Customize On-Demand

Swap Pricing

EffectiveDate	SpotLag	Maturity	Generator	PayRec	Notional	FixedRate	Spread	CreateSwap	CurveCollection	PV	ParRate	PV01	DV01
Thu, 24-May-18	2D	2Y	USD_OIS	PAY	1,000,000	1.00000%	0.0000	USD_OIS2Y@16:3	USDYC	28,153	2.4333%	196	192
Thu, 24-May-18	2D	5Y	USD_OIS	PAY	1,000,000	1.00000%	0.0000	USD_OIS5Y@28:3	USDYC	75,160	2.5994%	470	435
Thu, 24-May-18	2D	10Y	USD_OIS	PAY	1,000,000	1.00000%	0.0000	USD_OIS10Y@5:3	USDYC	147,500	2.6780%	879	777
Thu, 24-May-18	2D	2Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000	USD_3ML2Y@9:3	USDYC	35,132	2.8041%	195	194
Thu, 24-May-18	2D	5Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000	USD_3ML5Y@24:3	USDYC	93,501	3.0050%	466	441
Thu, 24-May-18	2D	10Y	USD_3ML	PAY	1,000,000	1.00000%	0.0000	USD_3ML10Y@21:3	USDYC	183,600	3.1050%	872	781
Thu, 24-May-18	2D	2Y	USD_6ML	PAY	1,000,000	1.00000%	0.0000	USD_6ML2Y@14:3	USDYC	38,261	2.9648%	195	204
Thu, 24-May-18	2D	5Y	USD_6ML	PAY	1,000,000	1.00000%	0.0000	USD_6ML5Y@15:3	USDYC	100,774	3.1610%	466	462
Thu, 24-May-18	2D	10Y	USD_6ML	PAY	1,000,000	1.00000%	0.0000	USD_6ML10Y@8:3	USDYC	198,007	3.2702%	872	801

Generator Customization

Display Swap Generator

LEGTYPE	Leg1:Fixed	Leg2:Float
LEVERAGE	1	1
FORECASTCURVE		USD6ML
COMPOUNDMETHOD	NONE	NONE
NOTIONALEXCHANGE	NONE	NONE
DISCOUNTCURVE	USDOIS	USDOIS
VALUATIONCURRENCY	USD	USD
CURRENCY	USD	USD
ACCRUALFREQUENCY	Semi-Annual	Semi-Annual
ACCRUALBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
ACCRUALCALENDAR	NYB:LNB	NYB:LNB
ACCRUALDAYCOUNT	30/360	ACT/360
PAYMENTFREQUENCY	Semi-Annual	Quarterly
PAYMENTBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
PAYMENTCALENDAR	NYB:LNB	NYB:LNB
PAYMENTLAG	0D	0D
FIRSTSTUBDATE		
LASTSTUBDATE		
ROLLDAY	NORMAL	NORMAL
STUBTYPE	ShortStart	ShortStart
FIXINGBUSINESSDAYADJUSTMENT		MOD_FOLLOWING
FIXINGCALENDAR		NYB:LNB
FIXINGLAG		0D
FIXINGADVANCEORARREAR		
FIRSTSTUBCURVEINDEX		NATURAL
LASTSTUBCURVEINDEX		NATURAL
#N/A	#N/A	#N/A
#N/A	#N/A	#N/A
#N/A	#N/A	#N/A
#N/A	#N/A	#N/A
#N/A	#N/A	#N/A
#N/A	#N/A	#N/A
#N/A	#N/A	#N/A
#N/A	#N/A	#N/A

Display Swap

LEGTYPE	Leg1:Fixed	Leg2:Float
FORECASTCURVE		USD6ML
COMPOUNDMETHOD	NONE	NONE
DISCOUNTCURVE	USDOIS	USDOIS
VALUATIONCURRENCY	USD	USD
CURRENCY	USD	USD
FIRSTFIXING		
LASTFIXING		
FIRSTSTUBCURVEINDEX		NATURAL
LASTSTUBCURVEINDEX		NATURAL
NOTIONAL	1000000	1000000
NOTIONALEXCHANGE	NONE	NONE
LEVERAGE	1	1
PAYRECEIVE	PAY	RECEIVE
EFFECTIVEDATE	43244	43244
MATURITYDATE	10Y	10Y
ACCRUALFREQUENCY	Semi-Annual	Semi-Annual
ACCRUALBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
ACCRUALCALENDAR	NYB:LNB	NYB:LNB
ACCRUALDAYCOUNT	30/360	ACT/360
PAYMENTFREQUENCY	Semi-Annual	Quarterly
PAYMENTBUSINESSDAYADJUSTMENT	MOD_FOLLOWING	MOD_FOLLOWING
PAYMENTCALENDAR	NYB:LNB	NYB:LNB
PAYMENTLAG	0D	0D
FIRSTSTUBDATE		
LASTSTUBDATE		
ROLLDAY	NORMAL	NORMAL
STUBTYPE	ShortStart	ShortStart
FIXINGBUSINESSDAYADJUSTMENT		MOD_FOLLOWING
FIXINGCALENDAR		NYB:LNB
FIXINGLAG		0D
FIXINGADVANCEORARREAR		
FIXEDRATE	0.01	
FLOATSPREAD		0



- Cashflow displays give a full break down of the Swap calculation

Display Cashflows

LEG1:FIXED Le

AccrualStart	AccrualEnd	AccrualDays	rualYearFracti	PaymentDate	Notional	ptionalExchanj	Leverage	FixedRate	Coupon	DiscountFactor	CouponPV
24-May-18	24-May-19	365	1.0000	24-May-19	-1,000,000	0	1.00	1.00000%	-10,000	1.003623	-10,036
24-May-19	25-May-20	367	1.0028	25-May-20	-1,000,000	0	1.00	1.00000%	-10,028	1.005269	-10,081
25-May-20	24-May-21	364	0.9972	24-May-21	-1,000,000	0	1.00	1.00000%	-9,972	1.003015	-10,002
24-May-21	24-May-22	365	1.0000	24-May-22	-1,000,000	0	1.00	1.00000%	-10,000	0.996996	-9,970
24-May-22	24-May-23	365	1.0000	24-May-23	-1,000,000	0	1.00	1.00000%	-10,000	0.988412	-9,884

LEG2:FLOAT Le

FixingDate	AccrualStart	AccrualEnd	AccrualDays	rualYearFracti	PaymentDate	Notional	ptionalExchanj	Leverage	FloatSpread	FloatRate	Coupon	DiscountFactor	CouponPV
24-May-18	24-May-18	43,430	186	0.5167	26-Nov-18	1,000,000	0	1.00	0.00	-0.31143%	-1,609	1.001883	-1,612
26-Nov-18	26-Nov-18	43,609	179	0.4972	24-May-19	1,000,000	0	1.00	0.00	-0.21800%	-1,084	1.003623	-1,088
24-May-19	24-May-19	43,794	185	0.5139	25-Nov-19	1,000,000	0	1.00	0.00	-0.08411%	-432	1.004913	-434
25-Nov-19	25-Nov-19	43,976	182	0.5056	25-May-20	1,000,000	0	1.00	0.00	0.11402%	576	1.005269	579
25-May-20	25-May-20	44,159	183	0.5083	24-Nov-20	1,000,000	0	1.00	0.00	0.30331%	1,542	1.004638	1,549
24-Nov-20	24-Nov-20	24-May-21	181	0.5028	24-May-21	1,000,000	0	1.00	0.00	0.51600%	2,594	1.003015	2,602
24-May-21	24-May-21	24-Nov-21	184	0.5111	24-Nov-21	1,000,000	0	1.00	0.00	0.71644%	3,662	1.000384	3,663
24-Nov-21	24-Nov-21	24-May-22	181	0.5028	24-May-22	1,000,000	0	1.00	0.00	0.88616%	4,455	0.996996	4,442
24-May-22	24-May-22	24-Nov-22	184	0.5111	24-Nov-22	1,000,000	0	1.00	0.00	1.02571%	5,243	0.992955	5,206
24-Nov-22	24-Nov-22	24-May-23	181	0.5028	24-May-23	1,000,000	0	1.00	0.00	1.15339%	5,799	0.988412	5,732

- Bucketed OIS & Libor Delta Risk for a EUR 1mm Eur6mL 3Y IRS

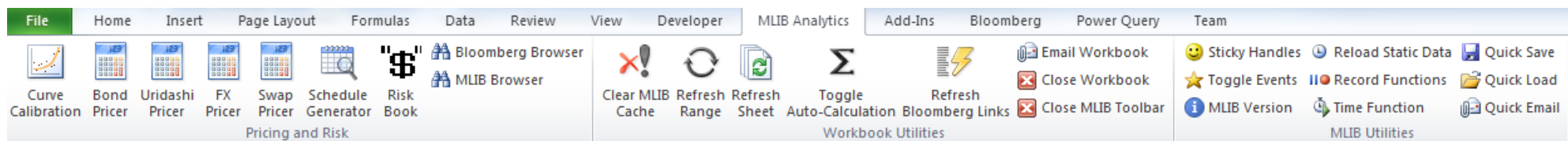
meLWOSwapDeltaLadder

EUR : Pillars	EUR : Total	EUR : Pillars	EUR : Total
EURYC_OIS_OUTRIGHTOISSWAP_ON	0	EURYC_STD_O_N	0
EURYC_OIS_OUTRIGHTOISSWAP_1W	0	EURYC_STD_T_N	0
EURYC_OIS_OUTRIGHTOISSWAP_2W	0	EURYC_STD_LIBOR_1M	0
EURYC_OIS_OUTRIGHTOISSWAP_3W	0	EURYC_STD_LIBOR_3M	0
EURYC_OIS_OUTRIGHTOISSWAP_1M	0	EURYC_STD_LIBOR_6M	4
EURYC_OIS_OUTRIGHTOISSWAP_2M	0	EURYC_STD_LIBOR_12M	0
EURYC_OIS_OUTRIGHTOISSWAP_3M	0	EURYC_STD_FRA_1X7	-5
EURYC_OIS_OUTRIGHTOISSWAP_4M	0	EURYC_STD_FRA_2X8	2
EURYC_OIS_OUTRIGHTOISSWAP_5M	-1	EURYC_STD_FRA_3X9	-1
EURYC_OIS_OUTRIGHTOISSWAP_6M	0	EURYC_STD_FRA_4X10	1
EURYC_OIS_OUTRIGHTOISSWAP_7M	0	EURYC_STD_FRA_5X11	0
EURYC_OIS_OUTRIGHTOISSWAP_8M	0	EURYC_STD_FRA_6X12	1
EURYC_OIS_OUTRIGHTOISSWAP_9M	0	EURYC_STD_FRA_7X13	-2
EURYC_OIS_OUTRIGHTOISSWAP_10M	0	EURYC_STD_FRA_8X14	1
EURYC_OIS_OUTRIGHTOISSWAP_11M	0	EURYC_STD_FRA_9X15	0
EURYC_OIS_OUTRIGHTOISSWAP_12M	10	EURYC_STD_FRA_10X16	1
EURYC_OIS_OUTRIGHTOISSWAP_18M	0	EURYC_STD_FRA_11X17	-1
EURYC_OIS_OUTRIGHTOISSWAP_2Y	21	EURYC_STD_6MLIBORSWAP_3Y	304
EURYC_OIS_OUTRIGHTOISSWAP_3Y	31	EURYC_STD_6MLIBORSWAP_4Y	-2
EURYC_OIS_OUTRIGHTOISSWAP_4Y	0	EURYC_STD_6MLIBORSWAP_5Y	1

- PV01 captures Libor Delta Risk only
- DV01 captures both OIS & Libor Delta Risk

Curve Calibration Toolkit

- Accessible from the MLIB Ribbon





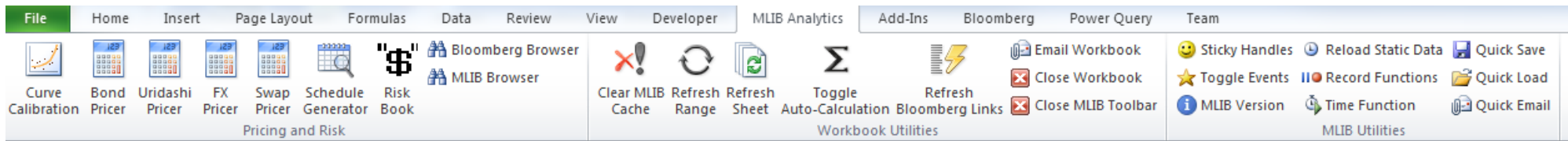


Refresh BGG Links

	Valuation Date Wed, 23-May-18	Enable Global Calibration FALSE	CSA Documentation ☐ Show / Hide	Enable CTD Curves FALSE	Allow Trade Updates FALSE	Enable Calibration Instruments FALSE
						
	(1) Market Data	(2) Native CSA	(3) USD CSA	(4) Non-Standard CSA	(5) Forward Charts	(6) Calibration Instruments
USD	☐ USD Market Data ✓0	☐ USD Curves ✓		☐ USD CSA Curves ✓	☐ USD Forwards	☐ USD Swaps
GBP	☐ GBP MarketData ✓0	☐ GBP Curves ✓	☐ GBP USD CSA ✓	☐ GBP CSA Curves ✓	☐ GBP Forwards	☐ GBP Swaps
EUR	☐ EUR MarketData ✓0	☐ EUR Curves ✓	☐ EUR USD CSA ✓	☐ EUR CSA Curves ✓	☐ EUR Forwards	☐ EUR Swaps
AUD	☐ AUD MarketData ✓0	☐ AUD Curves ✓	☐ AUD USD CSA ✓	☐ AUD CSA Curves ✓	☐ AUD Forwards	☐ AUD Swaps
JPY (JSCC)		☐ JPY JSCC Curves ✓			☐ JPY JSCC Fwds	☐ JPY JSCC Swaps
JPY (LCH)	☐ JPY MarketData ✓0	☐ JPY LCH Curves ✓	☐ JPY USD CSA ✓	☐ JPY CSA Curves ✓	☐ JPY LCH Fwds	☐ JPY LCH Swaps
JPY (TIBOR)		☐ JPY Tiber Curves ✓			☐ JPY Tiber Fwds	☐ JPY Tiber Swaps

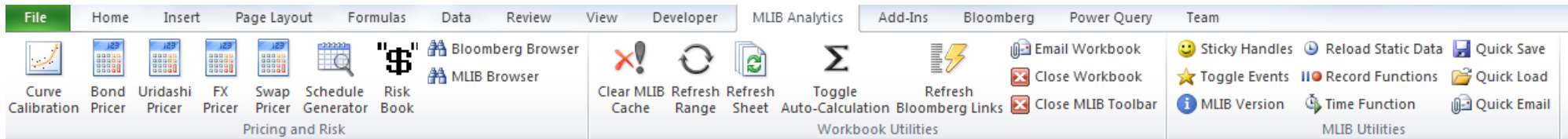
Bond & Uradashi Pricer

● Accessible from the MLIB Ribbon



Valuation Date	Settlement Date	1D	Bond Generator	US_TREASURY_TYPE1	YLD_YTM_ASK	BloombergYield	CleanPrice	DirtyPrice	Yield	ModDuration	DV01
Wed, 23-May-18	Thu, 24-May-18										
912828N8 Govt	US912828N894	US912828N894	1.375%	31-Jan-21	US912828N894@874:7	2.6659	96.6719	97.1011	2.6659216%	2.6013	252.58
912828B5 Govt	US912828B584	US912828B584	2.125%	31-Jan-21	US912828B584@873:7	2.6670	98.6016	99.2649	2.6669717%	2.5750	255.61
912828PX Govt	US912828PX26	US912828PX26	3.625%	15-Feb-21	US912828PX26@872:7	2.6563	102.5312	103.5126	2.6562562%	2.5669	265.71
912828P8 Govt	US912828P873	US912828P873	1.125%	28-Feb-21	US912828P873@871:7	2.6772	95.8828	96.1427	2.6772436%	2.6902	258.64
912828B9 Govt	US912828B907	US912828B907	2.000%	28-Feb-21	US912828B907@870:7	2.6710	98.2188	98.6807	2.6710222%	2.6594	262.43
912828Q3 Govt	US912828Q376	US912828Q376	1.250%	31-Mar-21	US912828Q376@869:7	2.6842	96.0859	96.2704	2.6842389%	2.7679	266.47
912828C5 Govt	US912828C574	US912828C574	2.250%	31-Mar-21	US912828C574@868:7	2.6789	98.8281	99.1601	2.6789003%	2.7330	271.01
912828Q7 Govt	US912828Q780	US912828Q780	1.375%	30-Apr-21	US912828Q780@867:7	2.6928	96.3047	96.3944	2.6928205%	2.8445	274.20
912828QN Govt	US912828QN35	US912828QN35	3.125%	15-May-21	US912828QN35@866:7	2.6958	101.2188	101.2952	2.6958398%	2.8252	286.18
912828R7 Govt	US912828R770	US912828R770	1.375%	31-May-21	US912828R770@865:7	2.7066	96.1641	96.8251	2.7066345%	2.9070	281.47
912828WN Govt	US912828WN60	US912828WN60	2.000%	31-May-21	US912828WN60@864:7	2.7051	97.9687	98.9303	2.7050686%	2.8766	284.58
912828S2 Govt	US912828S273	US912828S273	1.125%	30-Jun-21	US912828S273@863:7	2.7159	95.2969	95.7444	2.7158739%	3.0013	287.36
912828WR Govt	US912828WR74	US912828WR74	1.125%	30-Jun-21	US912828WR74@862:7	2.7140	98.2578	98.1031	2.7139737%	2.9524	292.59
912828S7 Govt	US912828S760	US912828S760	1.125%	31-Jul-21	US912828S760@861:7	2.7293	95.1328	95.4840	2.7292704%	3.0856	294.62
912828WY Govt	US912828WY26	US912828WY26	2.250%	31-Jul-21	US912828WY26@860:7	2.7311	98.5391	99.2414	2.7310523%	3.0307	300.77
912828RC Govt	US912828RC60	US912828RC60	2.125%	15-Aug-21	US912828RC60@859:7	2.7350	98.1250	98.7003	2.7350176%	3.0774	303.74
912828F2 Govt	US912828F677	US912828F677	1.125%	31-Aug-21	US912828F677@858:7	2.7418	94.9766	95.2364	2.7417691%	3.1654	301.46
912828D7 Govt	US912828D721	US912828D721	2.000%	31-Aug-21	US912828D721@857:7	2.7388	97.7031	98.1651	2.7387938%	3.1224	306.51
912828T3 Govt	US912828T347	US912828T347	1.125%	30-Sep-21	US912828T347@856:7	2.7477	94.8359	95.0019	2.7477156%	3.2476	308.53
912828F2 Govt	US912828F213	US912828F213	2.125%	30-Sep-21	US912828F213@855:7	2.7482	98.0156	98.3291	2.7481831%	3.1986	314.52
912828T6 Govt	US912828T677	US912828T677	1.250%	31-Oct-21	US912828T677@854:7	2.7518	95.1094	95.1909	2.7518208%	3.3225	316.27
912828F9 Govt	US912828F965	US912828F965	2.000%	31-Oct-21	US912828F965@853:7	2.7556	97.5391	97.6695	2.7556000%	3.2857	320.91
912828RR Govt	US912828RR30	US912828RR30	2.000%	15-Nov-21	US912828RR30@852:7	2.7496	97.5313	97.5802	2.7495727%	3.3260	324.55
912828U6 Govt	US912828U659	US912828U659	1.750%	30-Nov-21	US912828U659@851:7	2.7605	96.6328	97.4742	2.7605414%	3.3507	326.61
912828G5 Govt	US912828G534	US912828G534	1.875%	30-Nov-21	US912828G534@850:7	2.7493	97.0859	97.9874	2.7493473%	3.3430	327.57
912828G8 Govt	US912828G872	US912828G872	2.125%	31-Dec-21	US912828G872@849:7	2.7645	97.8203	98.6656	2.7645469%	3.4088	336.33
912828U8 Govt	US912828U816	US912828U816	2.000%	31-Dec-21	US912828U816@848:7	2.7681	97.3828	98.1784	2.7680586%	3.4165	335.43

• Accessible from the MLIB Ribbon



Illiquid Bond Price

100.890

Valuation Date (for Accrued Int)
Fri, 25-May-18

BOND SPECIFICATION	Proxy Bond	Uridashi Bond
ISIN	USJ46186BC59	XS1344710857
BondPrice	96.021	Uridashi Price
Notional	1,000,000	1,000,000
PayReceive	PAYER	PAYER
IssueDate	Thu 26-Mar-15	Fri 15-Jul-16
EffectiveDate	Fri 25-May-18	Fri 25-May-18
LastCouponDate	Mon 26-Mar-18	Mon 15-Jan-18
MaturityDate	Wed 26-Mar-25	Thu 15-Jan-26
CurveCollection	USDYC	AUDYC
ForecastCurve	USD3ML	AUD6ML
DiscountCurve	USDOIS	AUDOIS
isCleanPrice	TRUE	TRUE
FixedFrequency	SEMI-ANNUAL	SEMI-ANNUAL
FixedRate	3.20000%	3.92000%
FixedDayCount	30E/360	30E/360
FixedBusinessDayAdjustment	NO_CHANGE	MOD_FOLLOWING
FixedCalendar	NYB	SYB
FixedFirstStubDate		
FixedLastStubDate		
FixedRollDay	NORMAL	NORMAL
FixedStubType	ShortStart	ShortStart
FixedPaymentLag	0D	0D
CalculationType	Type1	

Create Asset Swap

PROXY_XCCY_ASSET_SWAP@2:7
Leg1:FixedBond USDYC
Leg2:Float AUDYC

Create Asset Swap

URIDASHI_ASSET_SWAP@3:7
AUDYC

Xccy Asset Swap Spread

96.04

Uridashi Price

100.890

Xccy Swap Properties

MTM	FALSE
ValuationCurrency	USD
FxRate	0.75000
NotionalFxResetLeg	

PROXY_XCCY_ASSET_SWAP

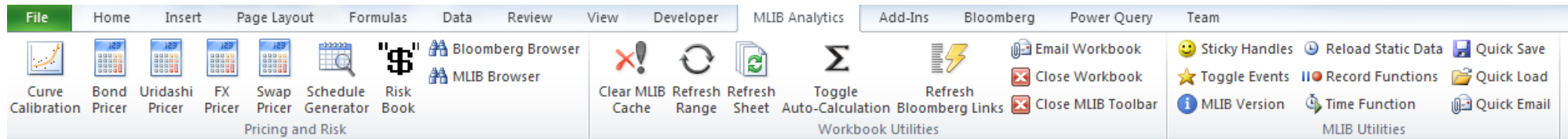
	Proxy Bond Leg	AUD Funding
LegType	Leg1:FixedBond	Leg2:Float
Notional	1,000,000	1,333,333
NotionalExchange	START_AND_END	START_AND_END
PayReceive	PAY	REC
ForecastCurve		AUD3ML
DiscountCurve	USDOIS	AUDDF_USDCSA
Currency	USD	AUD
Leverage	1	1
FixedRate	3.20%	
FirstStubCurveIndex		NATURAL
LastStubCurveIndex		NATURAL
FirstFixing		0.0171
LastFixing		
FloatSpread	0.00	0.00
EffectiveDate	Mon, 26-Mar-18	Mon, 15-Jan-18 *
MaturityDate	Wed, 26-Mar-25	Thu, 15-Jan-26
Frequency	SEMI-ANNUAL	QUARTERLY
Daycount	30E/360	ACT/365
FixingBusinessDayAdjustment	NO_CHANGE	NO_CHANGE
FixingCalendar	NYB	SYB
AccrualBusinessDayAdjustment	NO_CHANGE	NO_CHANGE
AccrualCalendar	NYB	SYB
PaymentBusinessDayAdjustment	NO_CHANGE	NO_CHANGE
PaymentCalendar	NYB	SYB

URIDASHI_ASSET_SWAP

	Leg1:FixedBond	Leg2:Float
Notional	1,333,333	1,333,333
NotionalExchange	NONE	NONE
PayReceive	PAY	REC
ForecastCurve		AUD3ML
DiscountCurve	AUDOIS	AUDOIS
Currency	AUD	AUD
Leverage	1	1
FixedRate	3.92%	
FirstStubCurveIndex		NATURAL
LastStubCurveIndex		NATURAL
FirstFixing		0.0171
LastFixing		
FloatSpread		90.81
EffectiveDate	Mon, 15-Jan-18	Mon, 15-Jan-18 *
MaturityDate	Thu, 15-Jan-26	Thu, 15-Jan-26
Frequency	SEMI-ANNUAL	QUARTERLY
Daycount	30E/360	ACT/360
FixingBusinessDayAdjustment	NO_CHANGE	NO_CHANGE
FixingCalendar	SYB	SYB
AccrualBusinessDayAdjustment	NO_CHANGE	NO_CHANGE
AccrualCalendar	SYB	SYB
PaymentBusinessDayAdjustment	NO_CHANGE	NO_CHANGE
PaymentCalendar	SYB	SYB

Swaps Pricer

- Accessible from the MLIB Ribbon



- Curve Support: EUR, USD, GBP and JPY JSCC & LCH

Valuation Date: Mon, 05-Jun-17

Bloomberg Access: TRUE

Pulse (Seconds): 15

Calc Mode: MANUAL

Use Forwards: FALSE
Curve has been set to build using Disco

DATA STREAMING

ENABLE Data Streaming

CALIBRATION

Step 1 Request Market Data

Step 2 Update Calibration Parameters

Step 3 Build Curves

CURVE BUILDING

Select Currency	Native CSA	CSA Curves
EUR	TRUE	Success
USD	TRUE	Success
GBP	TRUE	Success
JPY	TRUE	Success

Clear Curves & Object Cache

MARKET DATA

OIS CURVE Ready

	OIS SWAPS	OIS BASIS	LIBOR SWAPS
EUR	24	17	17
USD	14	12	12
GBP	20	0	0
JPY	20	0	0

SWAP CURVES Ready

	MM	FRAX3	FRAX6	FUT	SWAP
EUR	1	0	11	0	17
USD	1	0	0	12	10
GBP	1	0	8	0	16
JPY	1	0	5	0	16

BASIS CURVES Ready

	TENOR1	TENOR2	TENOR3	XCCY
EUR	16	14	16	15
USD	15	15	12	-
GBP	17	17	17	15
JPY	17	16	-	16

NAVIGATION

Show / Hide Calibration Instruments

Show / Hide Market Data

Show / Hide Curve Configurations

Show / Hide Curve Build & Charts

Products

- Asset Swaps, OIS, IRS and Tenor Basis, Xccy Swaps (incl. MtM)
- Variable Notional & Bespoke Swaps

SWAP PRICING	ASSET SWAP PRICING	TENOR BASIS SWAPS	VARIABLE NOTIONAL SWAPS	XCCY SWAPS
Show / Hide Swap Pricing Pages	Show / Hide Asset Swap Pricing Pages	Show / Hide Tenor Basis Pricing Pages	Show / Hide VNS Pricing Pages	Show / Hide Xccy Swaps
SWAP CASHFLOWS	ASSET SWAP CASHFLOWS	OVERNIGHT INDEXED SWAPS	STANDARD SWAP GENERATORS	CUSTOMIZED SWAPS
Show / Hide Swap Cashflow Pages	Show / Hide Asset Swap Cashflow Pages	Show / Hide OIS Pricing Pages	Show / Hide Standard Swap Pricing	Show / Hide Customized Swaps

Standard Swaps

- MLIB Supports Swap Templates or 'Generators'
- Simplified Trade Entry for Standard Swaps

Swaps Pricer: IRS Example

EUR Swaps 

PV **-299,129**

EffectiveDate	EffectiveDateOrLag	Notional	Currency	PayReceive	MaturityDate	FixedRate	FloatIndex	FloatSpread	CSA	ParRate	SwapPV	FixedPV	FloatPV	PV01	DV01
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	2Y	1.00000%	6ML	0.00	NOCSA	-0.15920%	23,295	20,096	3,199	201	204
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	3Y	1.00000%	6ML	0.00	NOCSA	-0.06430%	32,134	30,193	1,941	302	308
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	4Y	1.00000%	6ML	0.00	NOCSA	0.04610%	38,366	40,220	-1,854	402	411
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	5Y	1.00000%	6ML	0.00	NOCSA	0.16800%	41,795	50,235	-8,439	502	513
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	6Y	1.00000%	6ML	0.00	NOCSA	0.29360%	42,512	60,181	-17,669	602	613
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	7Y	1.00000%	6ML	0.00	NOCSA	0.42210%	40,470	70,029	-29,559	700	710
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	8Y	1.00000%	6ML	0.00	NOCSA	0.54970%	35,937	79,806	-43,870	798	805
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	9Y	1.00000%	6ML	0.00	NOCSA	0.67180%	29,327	89,357	-60,030	894	895
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	10Y	1.00000%	6ML	0.00	NOCSA	0.78480%	21,250	98,746	-77,496	987	994
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	11Y	1.00000%	6ML	0.00	NOCSA	0.88800%	12,095	107,989	-95,894	1,080	1,096
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	12Y	1.00000%	6ML	0.00	NOCSA	0.98050%	2,283	117,051	-114,769	1,171	1,198
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	15Y	1.00000%	6ML	0.00	NOCSA	1.19080%	-27,317	143,171	-170,488	1,432	1,365
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	20Y	1.00000%	6ML	0.00	NOCSA	1.37150%	-68,136	183,409	-251,545	1,834	1,698
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	25Y	1.00000%	6ML	0.00	NOCSA	1.44080%	-97,026	220,113	-317,139	2,201	2,001
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	30Y	1.00000%	6ML	0.00	NOCSA	1.47080%	-119,498	253,819	-373,317	2,538	2,278
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	40Y	1.00000%	6ML	0.00	NOCSA	1.47580%	-149,374	313,942	-463,316	3,139	2,779
Wed 7-Jun-17	2D	1,000,000	EUR	RECEIVER	50Y	1.00000%	6ML	0.00	NOCSA	1.42880%	-157,242	366,703	-523,945	3,667	3,273

Swaps Pricer: Custom Swap Example

CreateSwap EUR_CUSTOM_SWAP@45:1
DealName EUR_CUSTOM_SWAP
CurveCollection EURYC



Deal Information

	Leg1:Fixed	Leg2:Float
LegType	1,000,000	1,000,000
Notional	1,000,000	1,000,000
NotionalExchange	START_AND_END	START_AND_END
PayReceive	PAY	RECEIVE
EffectiveLag	2D	2D
EffectiveDate	Wed 7-Jun-17	Wed 7-Jun-17
MaturityDate	5Y	5Y
Frequency	Annual	Semi-Annual
ForecastCurve		EUR6ML
DiscountCurve	EUROIS	EUROIS
Currency	EUR	EUR
CompoundMethod	GEOMETRIC	GEOMETRIC
Leverage	1.00	1.00
FixedRate	1.00%	
FirstStubCurveIndex		NATURAL
LastStubCurveIndex		NATURAL
FirstFixing		
LastFixing		
FloatSpread		0.0000
Daycount	30/360	ACT/360
FixingBusinessDayAdjustment		MOD_FOLLOWING
FixingCalendar		TGT
AccrualBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
AccrualCalendar	TGT	TGT
PaymentBusinessDayAdjustment	MOD_FOLLOWING	MOD_FOLLOWING
PaymentCalendar	TGT	TGT
RollDay	NORMAL	NORMAL
FixingLag		0D
PaymentLag	0D	0D
FirstStubDate		
LastStubDate		
StubType	ShortStart	ShortStart

SwapName EUR_CUSTOM_SWAP@45:1
CurveCollection EURYC



Leg1:Fixed

LEG1:FIXED Leg 5

AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate	Notional	NotionalExchange	Leverage	FixedRate	Coupon	DiscountFactor	CouponPV
07-Jun-17	07-Jun-18	365	1.0000	07-Jun-17	1,000,000	1.00	1.00	0.0100	1,000,000	1.000018	1,000,018
07-Jun-18	07-Jun-19	365	1.0000	07-Jun-18	-1,000,000	0	1.00	0.0100	-10,000	1.003539	-10,035
07-Jun-19	08-Jun-20	367	1.0028	07-Jun-19	-1,000,000	0	1.00	0.0100	-10,000	1.006070	-10,061
08-Jun-20	07-Jun-21	364	0.9972	08-Jun-20	-1,000,000	0	1.00	0.0100	-10,028	1.006894	-10,097
07-Jun-21	07-Jun-22	365	1.0000	07-Jun-21	-1,000,000	0	1.00	0.0100	-9,972	1.005487	-10,027
				07-Jun-22	-1,000,000	-1,000,000	1.00	0.0100	-1,010,000	1.001486	-1,011,501

Leg2:Float

LEG2:FLOAT Leg 1

FixingDate	AccrualStart	AccrualEnd	AccrualDays	AccrualYearFractions	PaymentDate	Notional	NotionalExchange	Leverage	FloatSpread	FloatRate	Coupon	DiscountFactor	CouponPV
07-Jun-17	07-Jun-17	07-Dec-17	183	0.5083	07-Jun-17	-1,000,000	1.00	1.00	0.0000	-0.25600%	-1,000,000	1.000018	-1,000,018
07-Dec-17	07-Dec-17	07-Jun-18	182	0.5056	07-Dec-17	1,000,000	0	1.00	0.0000	-0.20700%	-1,301	1.001844	-1,304
07-Jun-18	07-Jun-18	07-Dec-18	183	0.5083	07-Jun-18	1,000,000	0	1.00	0.0000	-0.13000%	-661	1.005000	-664
07-Dec-18	07-Dec-18	07-Jun-19	182	0.5056	07-Dec-18	1,000,000	0	1.00	0.0000	-0.03563%	-180	1.006070	-181
07-Jun-19	07-Jun-19	09-Dec-19	185	0.5139	07-Jun-19	1,000,000	0	1.00	0.0000	0.06283%	323	1.006705	325
09-Dec-19	09-Dec-19	08-Jun-20	182	0.5056	09-Dec-19	1,000,000	0	1.00	0.0000	0.18326%	926	1.006894	933
08-Jun-20	08-Jun-20	07-Dec-20	182	0.5056	08-Jun-20	1,000,000	0	1.00	0.0000	0.30667%	1,550	1.006504	1,560
07-Dec-20	07-Dec-20	07-Jun-21	182	0.5056	07-Dec-20	1,000,000	0	1.00	0.0000	0.43969%	2,223	1.005487	2,235
07-Jun-21	07-Jun-21	07-Dec-21	183	0.5083	07-Jun-21	1,000,000	0	1.00	0.0000	0.58014%	2,949	1.003807	2,960
07-Dec-21	07-Dec-21	07-Jun-22	182	0.5056	07-Dec-21	1,000,000	1,000,000	1.00	0.0000	0.71598%	1,003,620	1.001486	1,005,111

The End